

Sentinel AI

Edge AI for Early Automotive Defect Detection

Catching systemic vehicle defects from the first few signals — before they become million-dollar recalls.

Vertical: Automotive — AI at the Edge for Vehicle Health

Track: Edge AI for Personalized & Connected Vehicles



Presentation Roadmap

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Who We Are & What We're Building

PROJECT OVERVIEW

EarlyWarn AI is an edge-deployed AI system that reads technician service notes and vehicle telemetry at the point of service, spotting emerging defect patterns from just a handful of cases — long before they escalate into fleet-wide recalls. Built for Tata Technologies InnoVent 2026, under the Automotive: Edge AI for Personalized & Connected Vehicles track.

OUR TEAM



AI / ML Lead

Model design, NLP & anomaly detection



Edge & Systems Engineer

On-device deployment & optimization



Data Engineer

Data pipelines & dealership integration



Product & Presentation

Problem framing, business case, delivery

TEAM NAME

Team Orvex

INSTITUTION

[Annasaheb Dange College Of Engineering And Technology Ashta]

WHY WE CARE ABOUT THIS PROBLEM

We started from a real, documented crisis at Tata Motors' own JLR operations — not a hypothetical — and wanted to build something that directly targets it with technology our team already knows well: NLP and applied ML.

Recalls Are Caught Too Late — After the Damage Is Done

Jaguar Land Rover — a Tata Motors subsidiary — has faced a wave of large-scale recalls: a corroding airbag clock-spring connector, a cracking front suspension knuckle, and a faulty DC-to-DC converter. In each case, the defect was only escalated into a formal recall after enough scattered dealer service reports and customer claims had piled up — by which point it had already spread across hundreds of thousands of vehicles.



The core issue: unstructured technician notes, diagnostic codes and sensor readings sit siloed across thousands of dealerships. Nobody connects the early, low-volume signals until a regulator forces the question.

WARRANTY COSTS AT JLR

6.6%

of sales — highest warranty cost ratio in 15 years

250,000+

vehicles — airbag corrosion recall

121,509

vehicles — suspension knuckle cracks

170,000

vehicles — DC-DC converter fault

A Large, Fast-Growing Problem Space

\$3.88B → \$7.08B

Automotive Warranty Management Software
Market, 2025 → 2033 (7.8% CAGR)

\$39.5B → \$62.6B

Global Auto Extended Warranty Market, 2025 →
2032 (6.8% CAGR) — the wider cost pool at risk

Sources: Mordor Intelligence / SkyQuest / 360iResearch industry market reports, 2025–26

WHY THIS OPPORTUNITY WILL ONLY GROW

- ✓ Rising vehicle software & electronics content creates more, and more complex, failure modes to track
- ✓ EV transition introduces new failure patterns (battery, power electronics) with limited historical data
- ✓ India's connected-vehicle cybersecurity & software rules phase in from Oct 2026, pushing OEMs toward proactive monitoring
- ✓ The same architecture extends beyond automotive — into aerospace MRO and industrial heavy-machinery warranty analytics



TARGET ADDRESSABLE SLICE

EarlyWarn AI targets the early-detection layer within warranty management software — the fastest-growing segment of a market already worth billions, directly reducing the cost pool shown on the left.

What We Aim to Do, and How



OBJECTIVE

Cut the time to detect an emerging vehicle defect from months of accumulating claims down to days — by reading the earliest signals at the point of service, entirely at the edge.

OUR APPROACH



1. Research & Data Study

Study historical recall/complaint datasets to learn what early defect signals actually look like



2. Model Design

Build NLP + anomaly-detection models sized to run on constrained edge hardware



3. Backtest & Validate

Validate against known past recalls — would the model have flagged them earlier?



4. Pilot & Iterate

Deploy to a small set of dealerships, gather feedback, refine thresholds

EarlyWarn AI: Spotting a Defect from the First Few Cases

An edge-deployed AI system that runs locally at dealer service centers, combining NLP on technician free-text notes with pattern analysis on vehicle diagnostic and telemetry data. It detects early, low-volume defect signatures — recognizing when a handful of seemingly unrelated service visits across different dealerships actually share a common root cause — and escalates only an anonymized pattern signal to engineering teams, long before claim volume is large enough to trigger a formal investigation.



NLP on Service Notes



On-Device Inference



Cross-Dealership Clustering



Privacy-Preserving
Escalation

FROM SIGNAL TO ALERT



Free-text + telemetry captured at dealership



Edge NLP + anomaly model scores each case

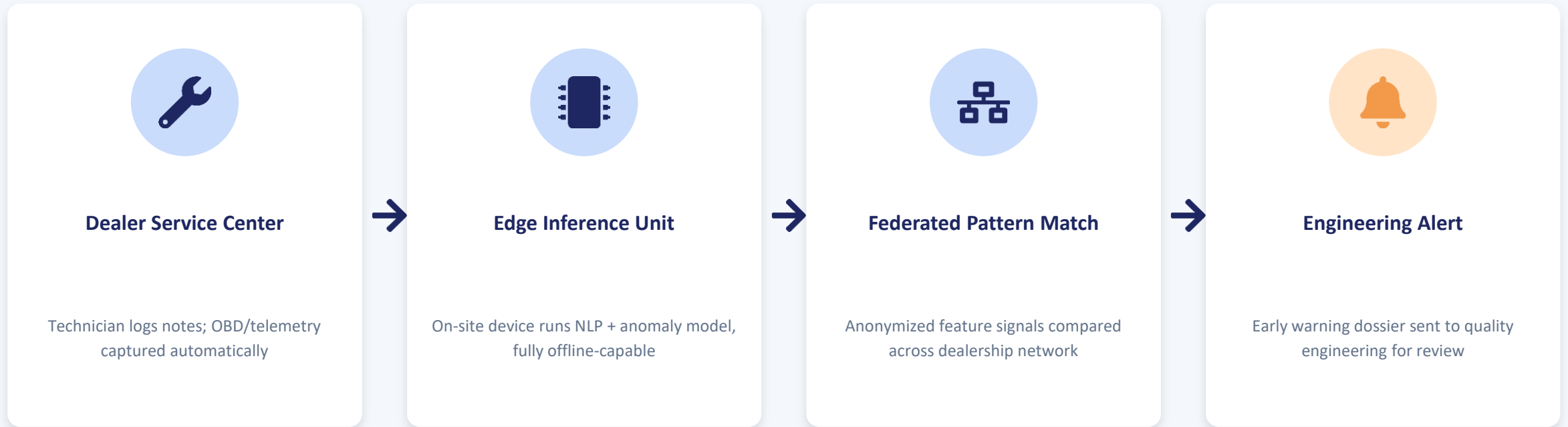


Similar cases clustered across locations



Engineering team alerted, early

How EarlyWarn AI Works, End to End



Closed feedback loop: confirmed findings retrain the edge model across the fleet, so detection sharpens with every case, not just at one dealership.

KEY FEATURES

What Makes EarlyWarn AI Different



NLP Defect Extraction

Understands free-text technician notes, not just structured fault codes



True Edge Inference

Quantized model runs on-site in kilobytes of memory, no cloud round-trip



Cross-Dealership Clustering

Connects 3-5 scattered cases into one emerging defect signature



Privacy by Design

Only anonymized pattern signals leave the dealership, not raw customer data



Days, Not Months

Cuts detection lag from a claims-driven investigation to a proactive flag



Works Offline

Fully functional at low-connectivity or remote dealer locations

Why This Hasn't Been Done This Way Before

TODAY

Reactive & Centralized

-  Service notes stay siloed per dealership, never cross-referenced
-  Central cloud analytics only run periodically, long after the fact
-  Needs a large claim volume before any pattern is even visible
-  Fails or delays further at low-connectivity dealer locations

WITH EARLYWARN AI

Proactive & Edge-Deployed

-  NLP reads technician notes locally, in real time, at every dealership
-  Clusters weak signals the moment 3-5 similar cases appear, anywhere
-  Flags an emerging defect signature within days, not months
-  Runs fully on-site — works even with zero or poor connectivity

Anticipated Challenges & How We Address Them



Cross-dealership data privacy

Federated design: only anonymized feature vectors leave a dealership, never raw text or customer data



Noisy, inconsistent technician notes

Domain-tuned NLP normalization pipeline handles shorthand, typos, and regional phrasing



Rare-event, imbalanced data

Framed as anomaly detection rather than supervised classification; augmented with historical recall datasets



Constrained edge hardware

Model quantization (TFLite/ONNX) keeps inference within kilobytes of memory and low latency



False-positive fatigue for engineers

Confidence scoring plus human-in-the-loop review to tune alert thresholds before scaling

Built for Real Edge Constraints



Data Layer

Dealer Management System logs, OBD-II/CAN telemetry, historical NHTSA/warranty claim datasets



Model Layer

DistilBERT-class NLP model (quantized) for note classification + Isolation Forest / lightweight autoencoder for telemetry anomaly scoring



Edge Runtime

TensorFlow Lite / ONNX Runtime Mobile on an in-dealership edge box (e.g. NVIDIA Jetson-class or Intel NUC-class hardware)



Coordination Layer

Lightweight federated signal aggregation — only feature vectors, never raw text, sync upstream when connectivity allows



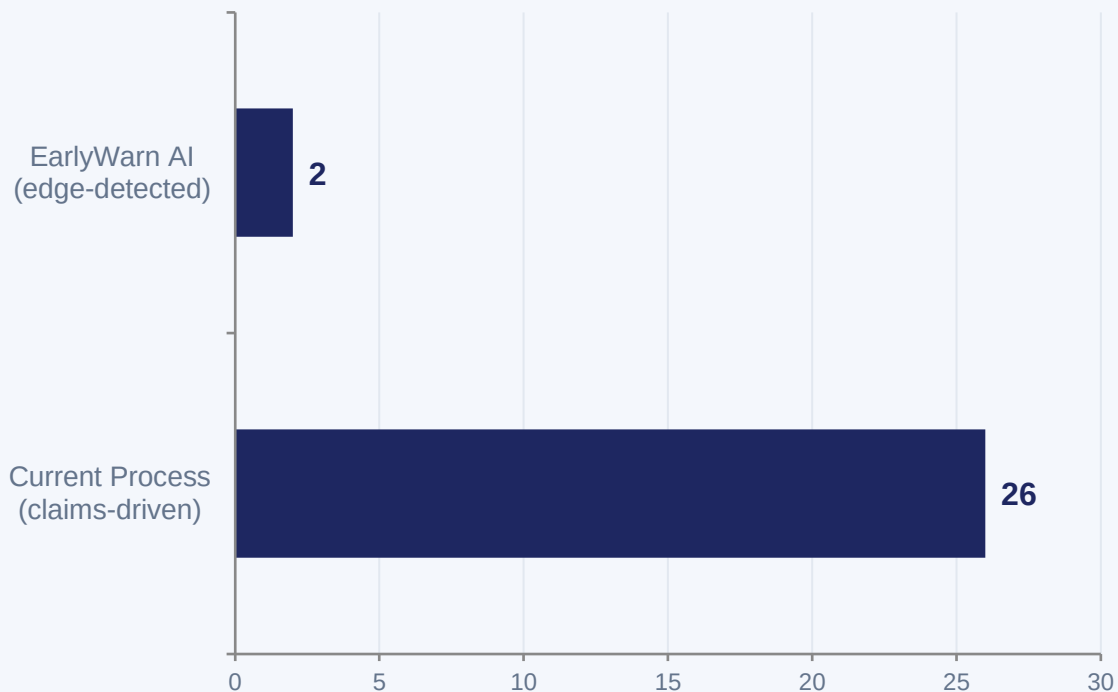
Application Layer

Engineering dashboard for cluster review, confidence scoring, and one-click escalation to quality teams

Target Outcomes, Validated Against Real History

These are projected outcomes from our validation methodology, to be confirmed through the PoC pilot described in the Project Plan.

Detection Timeline: Today vs. With EarlyWarn AI



Backtesting Methodology

Validate against the three real JLR recall cases: would our model have flagged the pattern earlier than the actual investigation trigger?



Warranty Spend Reduction

Targeting a measurable cut in the 6.6%-of-sales warranty cost ratio by containing defects before mass-recall scale



Safety-Relevant Signals, Sooner

Aiming to surface safety-relevant patterns in days, ahead of any regulator involvement

Where EarlyWarn AI Goes Next



Multilingual NLP

Extend note understanding to regional languages used across global dealership networks



Predictive Severity Scoring

Rank flagged clusters by estimated safety risk and cost impact, not just frequency



PLM / Quality System Integration

Direct integration with OEM product lifecycle & quality management platforms



Cross-Vertical Expansion

Adapt the same architecture to aerospace MRO and industrial heavy-machinery maintenance



Federated Learning at Scale

Continuously improve the shared model across regions without centralizing raw data



Technician Mobile App

Let field technicians flag suspected patterns instantly from a handheld device

From Pilot to Fleet-Wide Rollout





Thank You

Sentinel AI — catching the signal before it becomes a recall.

Questions & Discussion